

Question ID: 3e28a622

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function g is defined by $g(x) = \sqrt{x} + 300$. What is the value of $g(x)$ when $x = 81$?

Answer

- A. 9
- B. 300
- C. 309
- D. 381

Correct Answer: C

Rationale

Choice C is correct. It's given that function g is defined by $g(x) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g(81) = \sqrt{81} + 300$, which is equivalent to $g(81) = 9 + 300$, or $g(81) = 309$. Therefore, the value of $g(x)$ when $x = 81$ is 309.

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.